# A Multicenter Soft Supervised Classification Method for Modeling Spectral Diversity in Multispectral Remote Sensing Data

Jifa Guo and Shihong Du

***Abstract*— Due to the spectral diversity of objects within the same classes and the spectral similarity between different classes, classical fuzzy classification methods perform poorly in multispectral remote sensing image classification. The basic reason is that they often use a single spectral curve to represent a land cover type while ignoring spectral diversity characteristics. To solve this issue, this article proposes a novel multicenter supervised fuzzy classification (MCSFC) method for modeling spectral diversity in multispectral remote sensing data. Images are first split into clusters with similar sizes or volumes, namely, granularities, by a hierarchical clustering process. Second, the granularities are labeled by training samples. As a result, the centers of the granularities corresponding to one type of sample can represent the spectral diversity within one class and, thus, are treated as the multiple centers of a land cover type. The membership degree of each unlabeled sample belonging to any land cover type is determined by the shortest distance to the centers of each type, and the spectral diversity is considered in this stage. A comparison reveals that the proposed method clearly improves the classification performance of various single-center fuzzy semisupervised clustering (FSSC) methods. Two case studies indicate that the granularity volume parameter notably affects the classification performance, and the overall accuracy (OA) decreases with increasing granularity volume.**

***Index Terms*— Fuzzy semisupervised clustering (FSC), fuzzy set, fuzzy supervised classification, spectral diversity, spectral uncertainty.**

## I. Introduction

FUZZINESS is the inherent uncertainty that widely occurs in the geographic world [1]–[3], such as wetlands and forests, resulting in substantial challenges when processing and analyzing multispectral remote sensing data. Recently, fuzzy sets [4] have been widely applied in fuzzy land cover classification because they can express the gradation from one class to another [5]. In general, three types of methods have been developed. The first type involves unsupervised classification methods that require no prior knowledge. Fuzzy c-means (FCM) [6] clustering is a typical method of this type and is based on type-1 fuzzy sets [4]. Two parameters are required in this algorithm: the number of classes $N$ and fuzzifier $m$. However, different fuzzifiers $m$ may produce different FCM results; therefore, to address the uncertainty induced by the fuzzifier, Hwang and Rhee [7] developed the interval type-2 FCM (IT2FCM) clustering method via the interval type-2 fuzzy set (IT2 FS) [8], [9], which adopts two fuzzifiers to construct the footprint of uncertainty (FOU). The improved IT2FCM algorithm (IT2FCM*) [10] treats spectral curves as having a certain width to express the spectral uncertainty and then improves the classification accuracy. The second type is supervised methods that require prior knowledge to improve the classification accuracy. Ma *et al.* [11] reported that most supervised methods require 30%–50% of the total data as training samples. The c-fuzzy decision tree (CFDT) [12] is a typical method of this type with two stages. In the first stage, it applies the weighted FCM to build a decision tree (DT) by the training samples. In the prediction stage, it uses a hierarchical calculation of the membership degree to determine the type of unlabeled points. The third type is fuzzy semisupervised clustering (FSSC) [13], which was developed to improve the classification accuracy of FCM methods and reduce the difficulty of sample selection. Unlike supervised methods, FSSC methods require only limited labeled samples to improve performance. Several extensions of these methods have been proposed, including the semi-supervised FCM (ssFCM) [14], semisupervised kernel-based FCM (S2KFCM) [15], semisupervised fuzzy clustering with feature discrimination (SFFD) [16], semisupervised interval type-2 FCM using spatial information (SIIT2-FCM) [17], semisupervised kernel FCM in feature space (SKFCM-F), and semisupervised multiple-kernel FCM (SMKFCM) algorithms [18]. However, the improvement in the classification accuracy remains limited.

The inherent vagueness of geographic phenomena and the uncertainty introduced by sensors and data acquisition, processing, conversion, and transmission processes cause the well-known phenomenon of the same objects having different spectra [19], [20]. This phenomenon implies that, sometimes, the spectral values of a certain land cover type can vary substantially. Hence, multiple clusters with different shapes have often been observed in the multidimensional space of

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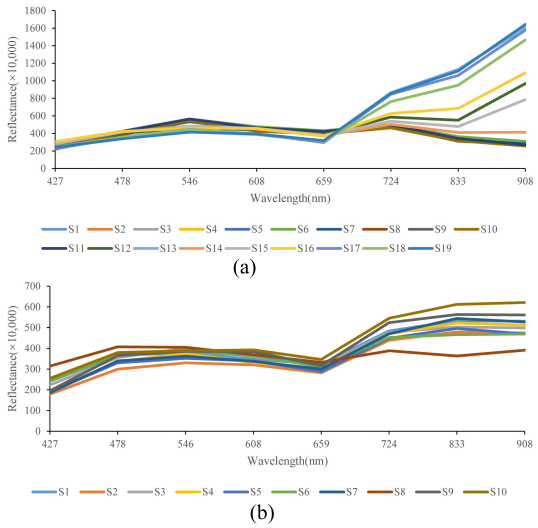


Fig. 1. Spectra of water bodies and shadows. The $x$-axis is the wavelength, and the $y$-axis represents the reflectance at a scale factor of 10 000. (a) Water bodies. (b) Shadows.

remote sensing data, leading to the inability of fuzzy unsupervised and semisupervised methods to suitably express the spectral diversity in multispectral remote sensing images. For example, the two subgraphs in Fig. 1 are the spectral curves of the same water body and shadow samples, which were manually selected from a WorldView-2 (WV-2) image. Due to the different chlorophyll, impurity, and water depth levels, the spectral reflectance of different water bodies may vary greatly, as different water bodies often correspond to multiple clusters in multidimensional space, as shown in Fig. 1(a). We found that the variations in the water samples are very large, and a single center does not suitably reflect the above characteristics. Therefore, the use of only one curve to represent various water samples is inadequate. In contrast, the other well-known uncertainty pertains to different objects with the same spectrum. In particular, the spectrum of a certain land cover type may be similar to that of another land cover type, which makes it difficult for the aforementioned methods to distinguish them. For example, in Fig. 1, the water body and shadow samples exhibit very similar spectra, while the differences between the water body samples are much larger than those between the water body and shadow samples. From the perspective of the methodology, a land cover type may contain some clusters, and if these clusters exhibit different shapes, volumes, or sizes, a single center may not represent the land cover type well. In other words, the use of a single center to represent the aggregation may induce uncertainties in the fuzzy classification results [7]. In addition, the overlap between clusters of different types further introduces uncertainty into the classification results [21].

Therefore, this complex intrinsic structure makes classification or feature extraction of remote sensing data very difficult [22],–[24]. Unfortunately, the abovementioned unsupervised and semisupervised fuzzy methods often ignore the complex intrinsic structure and apply a single point (a single- or interval-valued point) to represent the class centers, thus resulting in limited classification accuracy. In summary, the spectral diversity can be summarized into the following three levels: 1) land cover types may contain multiple clusters; 2) these clusters may have different sizes; and 3) some clusters may overlap each other. The FCM and FSSC methods cannot deal with these three levels of spectral diversity and have poor accuracy. The supervised methods assume that the training dataset should have a distribution similar to that of the full dataset. To ensure this assumption, a large number of training samples are required. However, selecting a large number of high-quality samples is expensive and difficult, especially when users are unfamiliar with the study area [16]. When the number and quality of samples are guaranteed, DT methods deal with the first level of spectral diversity very well because crisp DT methods, such as classification and regression trees (CART) [25] and C4.5 DTs [26], can deal with the second level, but they have a limited ability to deal with the third level. Because the type of geographic object is usually determined by several attributes, crisp DTs split a node into sons by a single attribute. A fuzzy DT, i.e., a CFDT, can deal with the first level but has limited ability to deal with the second and third levels of spectral diversity; a detailed analysis is provided later.

To handle these three levels of spectral diversity, first, a land cover type should be expressed as multiple clusters. For the second level, clusters of different sizes in multidimensional space should be identified. For the third level, overlapping clusters should be identified because they are responsible for producing classification uncertainty. From this point of view, we proposed a novel fuzzy supervised method, namely, the multicenter supervised fuzzy classification (MCSFC) method, to classify remote sensing data in this study. First, the full images are split into small granularities with similar sizes in spectral space by a hierarchical clustering process based on the FCM method. It is obvious that an original large and heterogeneous land cover type is split into a large number of granularities, while an original small or homogeneous land cover type consists of a small number of granularities. Second, labeled samples are employed to label and classify these granularities into three types: 1) pure granularities corresponding to only one type of sample, which are located in the inner clusters in spectral space (these samples are reliable for classification); 2) impure granularities corresponding to multiple types of samples, which are located at the transition zones between different types and are denoted by the boundary types; and 3) unlabeled granularities corresponding to no samples. Because the impure granularities are the overlapping clusters between different types, large uncertainties arise when using them as a classification reference. As a result, the impure granularities cannot be used as references for classification. The centers of all pure granularities are treated as representatives of all land cover types, and the spectral diversities of the various land cover types are then represented by these centers. The centers of each type are organized by the cover tree [27] to increase the retrieval speed of the center of the nearest neighbor. Third, the membership degree of each unlabeled sample belonging to any land cover type is determined by the nearest reliable reference cluster. We compare four single-center semisupervised methods (i.e., the ssFCM, SKFCM-F, SIIT2-FCM, and semisupervised adaptive interval type-2 FCM (SS-AIT2FCM)

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algorithms [28]) and three supervised methods (i.e., the CFDT, the online support vector machine (LaSVM) [29], and the random forest (RF) [30] methods). The comparison results reveal that the proposed method outperforms these methods. The main contributions of this study include the following.

1) A framework is proposed to model the spectral diversities of land covers by multiple centers, and these centers approximately describe the complex distribution of land covers in multidimensional space, which was selected from the leaves of the hierarchical tree by labeled samples.

2) Multiple centers were used in the classification process to solve the problems with single-center fuzzy classifications, which adopts a single center to express the spectral center of ground objects, and the classification accuracy of fuzzy classification was significantly improved after spectral diversity expression.

This article is organized as follows. In Section II, the standard FCM method is reviewed, and the problem that this study hopes to solve is described in detail. In Section III, the main ideology for constructing the algorithm proposed in this study is described. In Section IV, we present two experiments to examine the performance of the proposed method. Section V provides the conclusion.

## II. Prerequisites and Problem Statement

As mentioned before, the FCM method is suitable for modeling the fuzziness of land cover types. In this study, it was adopted to split an image into several granularities with similar sizes through a hierarchical clustering process. Hence, the samples in a given granularity exhibit a high spectral similarity. A brief introduction to the standard FCM method is provided in this section. In addition, we further state the issues to be addressed in this study.

### A. Fuzzy c-Means Algorithm

Let $X = \{x_{ij}, 1 \le i \le N, 1 \le j \le M\}$ be an $M$-dimensional sample set. In the standard FCM algorithm, the fuzzy objective function is

$$J_m = \sum_{k=1}^{C} \sum_{i=1}^{N} u_{ik}^m d_{ik} \tag{1}$$

where $N$ is the number of samples and $C$ is the number of clusters. It should be noted that the number of clusters is not necessarily equal to the number of classes. In addition, $m$ is the fuzzifier, $u_{ik}$ is the membership degree of sample $i$ belonging to cluster $k$, and $d_{ik}$ is the Euclidean or weighted Euclidean distance between sample $i$ and cluster center $k$. The cluster center $v_k = (v_{k1}, v_{k2}, v_{kj}, \ldots)$ and membership degree $u_{ik}$ are expressed as

$$v_{kj} = \sum_{i=1}^{N} u_{ik}^m x_{ij} \Big/ \sum_{i=1}^{N} u_{ij}^m \tag{2}$$

$$u_{ij} = 1 \Big/ \sum_{k=1}^{C} \left(d_{ij}/d_{ik}\right)^{2/(m-1)}. \tag{3}$$

The cluster centers and membership degrees are iteratively updated by (2) and (3), respectively.

### B. Problem Statement

The defaults of the above single-center classification methods are discussed in this section, and the effect of different cluster sizes on the classification accuracy is also assessed. The discriminatory power of the single-center classification methods is illustrated in Fig. 2(a). This dataset contains six clusters of different shapes and sizes. Suppose that this dataset should be classified into two classes, for example, water bodies and shadows, and the water body class contains clusters B, C, D, E, and F, while the shadow class contains only cluster A. As mentioned before, single-center fuzzy classification methods (such as the FSSC method) adopt one point to represent the cluster center, regardless of how many clusters the class contains. In Fig. 2(a), the centers of the two circles with equal radii are the centers of the two classes. The dotted line is the dividing line between the two classes. We observe that a single center leads to poor representativeness of the water body class and does not reflect that the water body class has multiple clusters. According to the principle of fuzzy classification that the unknown sample type is determined by the distance between this sample and the centers corresponding to all types, we can draw the separatrix line between these two types. We find that this line runs through clusters C and F, which means that, among the points in clusters C and F, the points on the right-hand side of the line are misclassified into class 1. This phenomenon leads to clusters C and F attaining a low classification accuracy. However, if the two centers of the two major clusters are adopted to represent the centers of the water body class, as shown in Fig. 2(b), these two centers describe, if not completely, the aggregation diversity characteristics of this type. According to the principle of fuzzy classification, in this case, the unknown sample type is determined by the distance between this sample and the nearest centers corresponding to all types. In Fig. 2(b), the types of samples in clusters C and F are determined by centers 1 and 2, respectively, and the separatrix line indicates that the classification accuracy of clusters C and F is greatly improved. This is closer to the goal of clustering, namely, achieving small within-class scatter but large between-class separation [31]. Therefore, single-center classification methods cannot represent the first level of spectral diversity, leading to poor performance in remote sensing data analysis.

The CFDT applies the FCM method to construct the DT and produce clusters with different sizes for different land cover types. However, in its prediction stage, it experiences similar problems to those in the FCM and FSSC methods, as shown in Fig. 2(c). These fuzzy methods produce clusters with different sizes because they apply the relative distance to determine the membership degree, which causes misjudgment of the points near the separatrix line [7]. As a result, it cannot effectively deal with the second level of spectral diversity. Another fault of the CFDT is that it uses impure clusters as classification references. The CFDT regards the majority among the boundary clusters as the representative type of this type of cluster in the soft DT. For example, a leaf in the DT may have three types of samples: one sample of type A, 12 samples of type B, and two samples of type C. Hence, a leaf may be labeled B. Sometimes, when the number of

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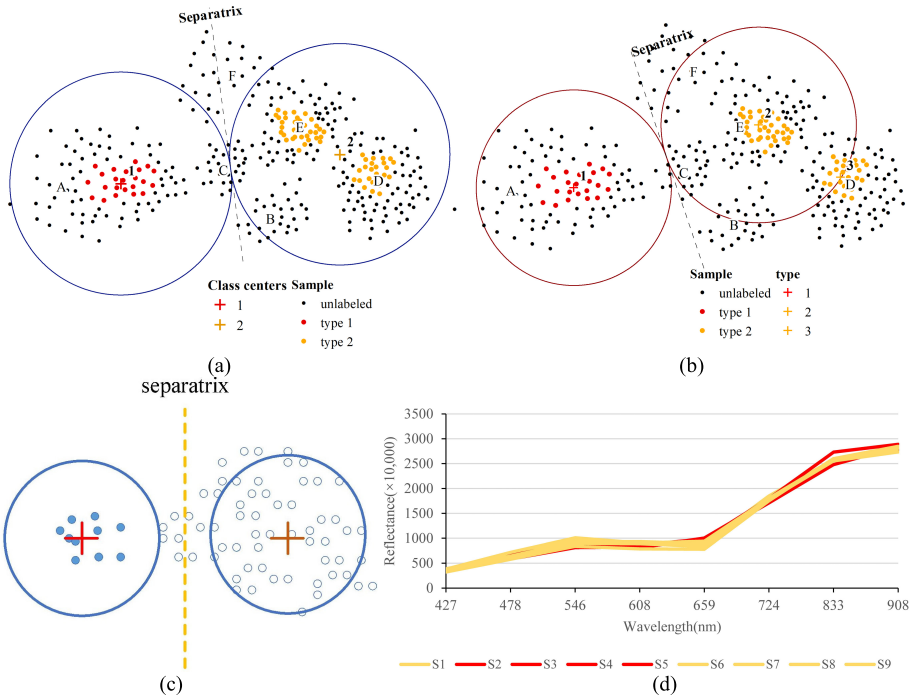


Fig. 2. (a) Single-center classification strategy: the centers of the two circles with equal radii are the centers of the two classes. The dotted line is the dividing line between the two classes. (b) In the multicenter classification strategy, two centers are employed to represent the centers of type 2. (c) Different cluster sizes lead to misjudgment. (d) Spectra of certain construction samples are very similar to those of certain bare soil samples; S1 and S6–S9 are the spectra of five bare soil samples, and S2–S5 are the spectra of four construction samples. The $x$-axis is the wavelength, and the $y$-axis represents the reflectance for a scale factor of 10 000.

samples of each type is similar, it is difficult to judge the main types. Fig. 2(d) reflects this case, and it contains some spectral curves of samples in an impure leaf, which are generated by the CFDT. Therefore, this type of leaf may result in a large classification error in the CFDT, limiting the ability of the CFDT to handle the third level of diversity.

## III. Methodology

In this section, we illustrate the MCSFC method. In the proposed method, dataset $X$ is adopted to directly generate clusters through a hierarchical clustering process without labels. Hence, samples with similar spectral attributes occur in a given cluster. This hierarchical clustering process generates a cluster tree, and the labeled samples are then used to mark and classify the leaves into three types: pure, impure, and unlabeled leaves. The centers of the pure leaves are treated as the multiple centers of the land cover types, and the membership degree of each unlabeled sample among the impure and unlabeled leaves is determined by the shortest distance between this sample and the multiple centers. The procedures of the MCSFC method are shown in Fig. 3.

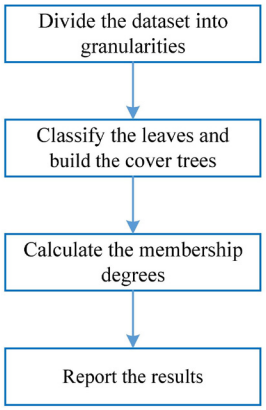


Fig. 3. Procedure of the MCSFC method.

### A. Dividing the Dataset Into Granularities by a Hierarchical Clustering Process

To model the first and second levels of diversity, the full dataset is split into small granularities with similar sizes in spectral space in this step. A hierarchical clustering process based on the FCM algorithm with the weighted Euclidean distance is adopted to generate a hierarchical tree. This division process is similar to the hierarchical FCM algorithm [32], which is realized by repeatedly running the conditional FCM algorithm [33]. The difference is that the membership degrees in the parent node are treated as a condition of child nodes in the hierarchical FCM algorithm. It should be noted that, although the IT2FCM algorithm has been indicated to be more effective than the standard FCM, it is not employed in this study because it is time-consuming. In our test, the division process was excessively long and lost its practical value; thus, it is unsuitable for this process in this study. After the division step, a hierarchical tree can be achieved, as shown in Fig. 4. The red and blue circles represent intermediate and leaf nodes, respectively.

In this division process, we first use the FCM algorithm to split the full dataset $X$ into $C$ leaves, and the weighted

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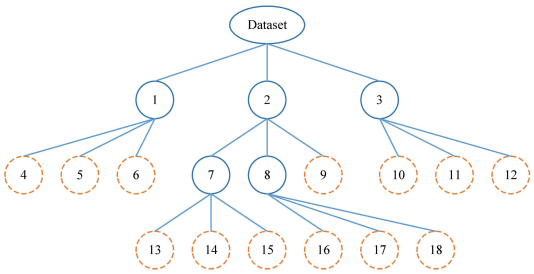


Fig. 4. Hierarchical cluster tree.

Euclidean distance is determined as follows:

$$d_{ik} = \sum_{j=1}^{M} \left(x_{ij} - v_{kj}\right)^2 \Big/ \sigma_j^2 \tag{4}$$

where $d_{ik}$ is the weighted Euclidean distance between sample $x_i$ and center $v_k$, and $\sigma_j$ is the standard deviation of dimension $j$ of the dataset. Many distance definitions are available for remote sensing data analysis [34]; thus, we tested the performances of the Euclidean distance, the weighted Euclidean distance, the spectral angle metric (SAM), the spectral information divergence (SID), the SAM-SID mixed measure, and the Jeffries–Matusita-spectral angle mapper (JM-SAM) [35], [36]. The weighted Euclidean distance is selected because it has the highest classification accuracy and efficiency. After applying the FCM algorithm to the full dataset, we obtain $C$ clusters, and the full dataset is then split as follows:

$$x_k = \left\{ x_i \middle| u_{ik} = \max_{k=1,2,\ldots,C} (u_{ik}) \right\}. \tag{5}$$

These $C$ clusters are treated as the leaves of a parent node. We take the number of samples in a cluster as the termination condition. That is, if the sample number of a leaf is larger than the number threshold, the FCM is repeatedly applied to this leaf with the same parameters as the first node of the tree. Otherwise, it is not necessary to split the leaf. The process is illustrated by Fig. 4. First, the full data are split into three clusters, labeled 1, 2, and 3. Because the number of samples is greater than the threshold, all three clusters should be split further. Thus, node 1 is split into nodes 4, 5, and 6, and nodes 2 and 3 are also divided accordingly. However, the numbers of samples in nodes 4, 5, and 6 are less than the threshold; thus, they cannot be divided further. The splitting operation should be run repeatedly for nodes whose numbers of samples are greater than the threshold until the division termination criterion is reached.

After this division process, all the samples in the dataset are clustered into small leaves, the samples in a leaf have highly similar spectral attributes, and the number of samples of all leaves in the hierarchical tree is smaller than the threshold. We refer to these leaves as granularities. In addition to the number of samples, the maximum value of the difference between the supremum and infimum of each dimension of the data samples can also be adopted as the threshold. In particular, this value can be treated as the maximum size of a cluster, and in our experiments, these two kinds of thresholds yield consistent results. It is clear that the parameters of the FCM algorithm and the number threshold greatly influence the structure of the hierarchical tree; that is, different parameters may produce different hierarchical trees and data partitions. In this study, the two most influential parameters are the fuzzifier $m$ and the threshold. The effect of $m$ in the FCM algorithm has been widely discussed and, thus, is not discussed in this study. For the threshold, we discuss its effects in the next section. For parameter $C$, although different $C$ values can produce different trees, the final classification accuracy is very close in our test; thus, it is not discussed in depth. Regarding the maximum number of iterations and error tolerance in the FCM algorithm, it is well known that a larger number of iterations and a smaller error tolerance produce better results; thus, further elaboration is not provided. Consequently, in this study, a large number of iterations and a small error tolerance are adopted for the FCM algorithm.

Another difference between our method and the hierarchical FCM algorithm is the termination condition. The termination criterion in the hierarchical FCM algorithm was defined as the approximation ability of each node center. If the distribution of the samples in a cluster is symmetric, even if the size of a cluster is large, the termination criterion may have a small value. Therefore, the size or volume of the leaves produced by the hierarchical FCM tree may vary greatly, which affects the classification accuracy, as discussed in Section II-B. In our method, the sizes of clusters are very similar, which reduces the uncertainty caused by the cluster sizes. Therefore, in our experiments, the hierarchical FCM tree attains a lower accuracy than the proposed method. The proposed process is also different from those of the CART and CFDT methods. These two methods employ a training sample set to build the DT. The CART method uses one attribute to split a node during the DT-building process by a certain criterion, such as the entropy or Gini index. This study and the CFDT apply all attributes to split the nodes during the tree-building process; this development approach promotes more compact trees and a versatile partition geometry of the feature space [12]. Moreover, since the CFDT uses the variability or structurability index measure as the node splitting criterion, the CFDT may, therefore, produce clusters with very large volume differences, as discussed in Section II-B.

### *B. Classifying the Leaves and Building the Cover Trees for Each Type*

In this step, all leaves of the hierarchical tree are selected and then classified into three types: 1) pure labeled leaves with labeled samples of a single type, with the concomitant condition that the type of labeled sample in a leaf is singular and the number of labeled samples is larger than 4 [37], [38]; 2) impure labeled leaves with labeled samples of different types; and 3) unlabeled leaves. Fig. 5 illustrates the pure, boundary, and unlabeled leaves and the cover trees of each land cover type, where clusters 1–4 are the pure clusters of type 1, clusters 6–8 are the pure clusters of type 2, and clusters 9 and 10 are two unlabeled leaves. In this study, these kinds of leaves are denoted as interior leaves and used to represent

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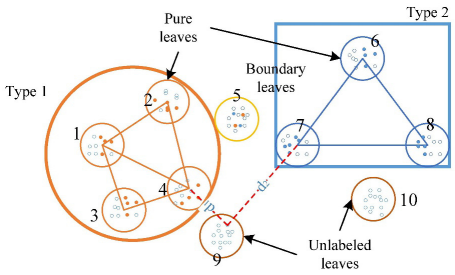


Fig. 5. Pure labeled leaves, boundary leaves, unlabeled leaves, and the cover trees of each land cover type.

the spectral diversity of types and provide a reliable reference for classification. Cluster 5 is an impure labeled leaf because it contains labeled samples belonging to both types 1 and 2, and these kinds of samples occur in the transition zone of the two types. Intuitively, the wider the transition zone is, the harder it is to determine the dividing line between the two types, and we denote these kinds of leaves as boundary leaves. If the number of labeled samples of certain types in a cluster is similar, it becomes difficult to distinguish these types, as shown in Fig. 2(d), because all samples in this cluster exhibit similar spectral attributes. Therefore, this type of node is representative of uncertainty, i.e., the phenomenon of different objects with the same spectrum. That is, this type of leaf has great classification uncertainty and cannot be used as a reference for classification. However, the existing FSSC and DT methods cannot identify and exclude such samples. Therefore, accurately identifying interior and boundary leaves is very important for modeling spectral diversity and classification.

In this study, the centers of all pure labeled leaves are treated as multiple centers of the land cover types. In Fig. 5, four centers are adopted to represent type 1, three centers are used to represent type 2, and these centers can well represent the arbitrary shapes of the data distribution and the multiple aggregations of one type. In other words, many centers may be obtained for any given land cover type to express the spectral diversity of various geographic objects and phenomena. However, the size of each cluster is quite similar in this study, avoiding the classification uncertainty caused by the large difference in cluster volume.

It should be noted that the size threshold controls the ability to measure spectral diversity. A larger value leads to a smaller number of large granularities, which has two effects. The first effect is that it causes large differences between the granularity sizes, which further increases the uncertainty. The second effect is that the size of the boundary leaves increases as the threshold value increases. Therefore, a smaller number of pure leaves are produced, and it is difficult to represent the spectral diversity with a small number of centers. On the other hand, the boundary zones between land cover types constructed by impure leaves increase; at the same time, the distances between the centers of reliable granularities as classification references and the boundary zones increase, which also increases the classification uncertainty and reduces the classification accuracy. In contrast, a relatively small threshold can produce many small pure granularities, which can help to disclose the data distribution of types more precisely and identify the uncertainty boundary more accurately. However, a threshold that is too small causes the division results to be very fragmented and, thus, cannot help to identify the uncertain region. In the extreme case where the threshold is 1, which means that the data are not divided, we cannot recognize the transition zones or boundary zones between the types and cannot distinguish between reliable and unreliable reference samples for classification.

### C. Calculating the Membership Degrees of the Unlabeled Samples

As mentioned before, the membership degree of an unlabeled sample belonging to any land cover type is determined by the shortest distance between this sample and the multiple centers. Therefore, for an unlabeled sample, to quickly determine its nearest center of a certain land cover type, a cover tree [27] is employed to manage the centers of the pure leaves. In our experiments, the distance metric used in the cover tree has some impact on the results, and the weighted Euclidean distance with a weight of $1/\sigma_j^2 (1 \le j \le M)$ is selected for the cover tree. In this study, an open-source Java library, namely, the Statistical Machine Intelligence and Learning Engine (Smile) [39], is adopted to build the cover trees. That is, for each land cover type, we use its central points to build a cover tree; if we want the dataset to be classified into $H$ types, $H$ cover trees need to be built.

As discussed before, the membership degree of an unlabeled sample belonging to a land cover type is determined by the shortest distance between this sample and the nearest center of this type. In Fig. 5, for an unlabeled sample in cluster 9, because it is located between type 1 and 2 and its nearest center to type 1 in the cover tree is the center of cluster 4, the shortest distance is $d_1$. Similarly, the shortest distance between this sample and type 2 is $d_2$. Finally, the membership degree of an unlabeled sample belonging to types 1 and 2 is determined by $d_1$ and $d_2$, respectively.

Suppose that the data are to be classified into $H$ types. For the unlabeled samples in the pure labeled leaves, because the samples in a cluster exhibit a high spectral similarity, the simplest way to address them is to apply the labeled type as their type and set the membership degree of this type to 1 and that of the other types to 0. For the unlabeled samples in the other leaves, the membership degrees are determined by

$$u_{ij} = 1 \Big/ \sum_{k=1}^{H} \left(d'_{ij}/d'_{ik}\right)^{2/(m-1)} \tag{6}$$

where $u_{ij}$ is the membership degree of sample $i$ belonging to type $j$, and $d'_{ij}$ and $d'_{ik}$ are the shortest distances from sample $i$ to the center of types $j$ and $k$, respectively.

Similar to the other pixel-based methods, (6) does not consider spatial information, leading to obvious salt-and-pepper noise. To solve this problem, the mean smoothing method was applied to consider the neighbor membership degree. The principle of the maximum membership degree was used

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to determine the type of unlabeled samples, which can be expressed as $u_{ih} > \max_{\substack{h \neq j \\ j=1,2,\ldots,H}}(u_{ij})$.

### D. Computational Complexity Analysis

The computational complexity of the proposed method consists of three parts. The first is the complexity of the division process. The computational complexity of the first level of the hierarchical tree is equal to that of the standard FCM algorithm, that is, $O(\mathrm{CNM})$ [40], where $C$ is the cluster number, $N$ is the number of samples, and $M$ is the number of attributes. To estimate the overall complexity, we assume that each child of a node has the same number of samples; thus, in the second level, the number of samples in each leaf is $N/C$. That is, the computational complexity of each leaf is $O(C(N/C)M)$; as a result, the computational complexity of this level is $O(C(N/C)M) \times C = O(\mathrm{CNM})$. For the third level, the number of samples at each leaf is $N/C^2$, leading to a computational complexity at this level of $O(C(N/C^2)M) \times C^2 = O(\mathrm{CNM})$. Therefore, each level has the same computational complexity. The number of samples at a leaf in level $l$ is $N/C^{l-1}$, the termination condition is that $N/C^{l-1}$ is less than the threshold value Thr, and the maximum number of levels can be estimated by $L = \mathrm{Int}(\log_C N/\mathrm{Thr} + 1)$. The function Int() means taking the smallest integer greater than a floating number. Therefore, the overall computational complexity of the division process is $L \times O(\mathrm{CNM})$.

For the second part, the computational complexity of constructing the cover tree is $O(\delta^6 N' \ln(N'))$ [27], where $N'$ is the number of centers for constructing the cover trees and $\delta$ is a bounded expansion constant. However, because the number of centers is far less than that of samples, i.e., $N' \ll N$, the time consumed in this part is far less than that of the first part. In the membership degree calculation stage, each unlabeled sample contains two steps: nearest neighbor query and membership degree calculation. Similarly, because $N' \ll N$, the computational complexity of this part is $O(N)$.

In summary, the computational complexity of the MCSFC method should be $L \times O(\mathrm{CNd})$.

## IV. Experiments

In this section, we consider two cases to test the reliability of the proposed method for different datasets. The first case involves the WV-2 dataset. Our method is compared to the three aforementioned supervised classification methods, namely, CFDT, LaSVM, and RF, and to the above four semisupervised classification methods, namely, the ssFCM, SKFCM-F, SIIT2-FCM, and SS-AIT2FCM algorithms, by using the confusion matrix and kappa coefficient. We examine the influences of different fuzzifier values and granularity sizes on the number of granularities and classification accuracy. The second case pertains to the Sentinel-2 dataset, and we conduct a similar examination to that in the first case. The scenarios covered by the two datasets are also different. The first dataset mainly covers an urban landscape, while the second dataset largely covers a wetland landscape. Therefore, through these two cases, the applicability of the proposed method under different scenarios can be validated.

Fig. 6. Natural color composite image of the WV-2 dataset using bands 5, 3, and 2.

### A. WorldView-2 Dataset

In this case, the multispectral WV-2 image is adopted to test the classification performance of the proposed method. The WV-2 image contains a panchromatic band with a spatial resolution of 46 cm, and the spatial resolution of eight multispectral bands (coastal blue, blue, green, yellow, red, red-edge, near-infrared 1 (NIR1), and NIR2) is 1.8 m. The image was acquired on September 13, 2015. Eight multispectral bands were selected and resampled at a 2-m spatial resolution, and a subset with a size of 633 × 997 was extracted. It covers the campus of Tianjin Normal University, which is located in southwestern Tianjin city, a metropolis on the northern coast of mainland China. The land cover types mainly include buildings, wetlands, trees, grasslands, bare soil, and concrete roads. The reason for choosing this region as the study area is primarily due to the convenience of validation. The radiometric and atmospheric corrections were performed in ENVI 5.3. The natural color composite image using bands 2, 3, and 5 is shown in Fig. 6. The testing samples were collected by visual interpretation and investigated in ArcGIS 10.5. The training samples were extracted from the test dataset, and samples of each class should contain the main subclass. For example, the test samples of water should contain samples in deep water, shallow water, water with chlorophyll, and water with submerged plants. The training and test samples are shown in Fig. 7, and the training dataset covers approximately 8.7% of the study area and is listed in Table I.

*1) Spectral Diversity and Classification Results:* In the proposed MCSFC method, the first step is to split the image into granularities by using the method described in Section III-A. Parameter $C$ of the FCM algorithm is set to 3, the fuzzifier is set to $m = 2.0$, the maximum number of iterations is set to 100, and the error tolerance is 0.001. The threshold of

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES IN DATASET 1

| Type | Number of training samples | Number of test samples |
|---|---|---|
| Water | 26,784 | 123,998 |
| Shadows | 1733 | 18,604 |
| Construction | 14,784 | 83,547 |
| Bare soil | 1980 | 3055 |
| Woodland | 8354 | 57,066 |
| Grassland | 4021 | 39,337 |

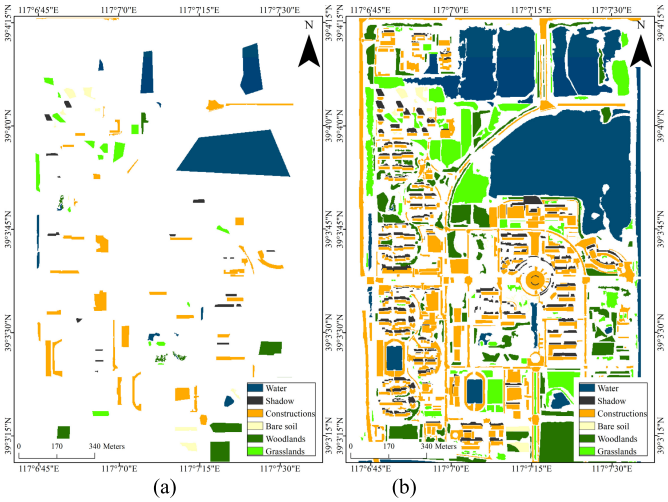


Fig. 7. (a) Training and (b) test samples for case 1.

the leaf size is set to 100. After the hierarchical clustering process, 13 521 leaves are produced, as shown in Fig. 8(a). The second step is to divide these leaves into 3057 pure labeled leaves, 69 impure labeled leaves, and 10 395 unlabeled leaves by the method described in Section III-B. In addition, the numbers of pure water, shadows, construction, bare soil, woodland, and grasslands levels are 1203, 116, 911, 107, 470, and 250, respectively. As shown in Fig. 8(b), the pure leaves reflect the spectral diversity of the six types. For example, pure leaves of water include different water subtypes, such as deep water, shaded water, or water with high chlorophyll content. In this study, the centers of all these pure leaves are adopted as classification references. Therefore, compared with the single-center methods, the classification reliability is improved. Some labeled pixels in Fig. 7(a) are removed in Fig. 8(b) because they occur in the impure leaves and are not used as a classification reference in this study. This also helps to enhance the classification accuracy.

In the third step, we establish six cover trees by employing the centers of the pure labeled leaves of each type, and then, the membership degree is determined by the method presented in Section III-C. The classification results are shown in Fig. 9. The confusion matrix and accuracy are listed in Table II. The producer's accuracies (PAs) and user's accuracies (UAs) of all the land cover types are above 80%, and the construction areas attain the highest accuracy of 98.71%. The water and shadow

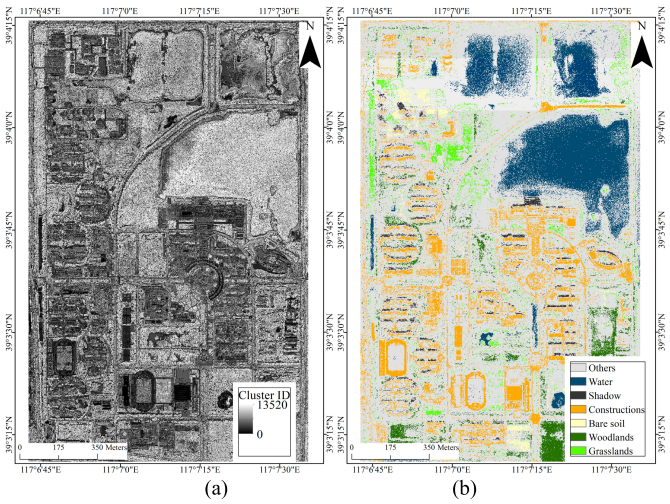


Fig. 8. (a) Full partition of image 1. (b) Spatial distribution of the pure granularities corresponding to the six types in image 1.

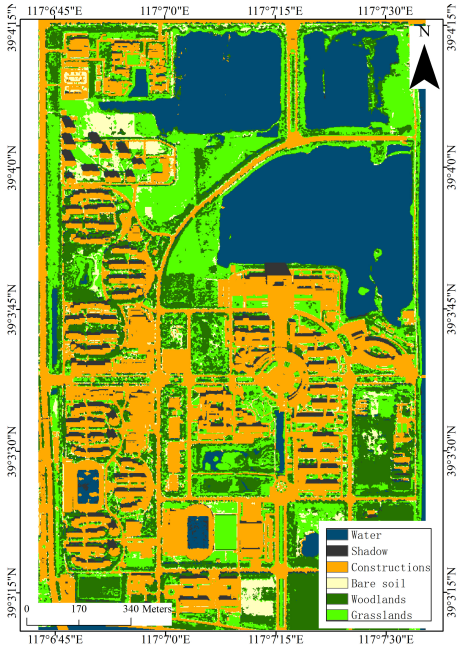


Fig. 9. Classification result of the WV-2 dataset by the MCSFC method.

types can clearly be distinguished with accuracies of 98.04% and 83.23%, respectively.

*2) Accuracy Comparison:* This section compares the MCSFC method with the three supervised methods: the CFDT, LaSVM, and RF algorithms. The parameter values of the FCM algorithm in the CFDT are the same as those in the MCSFC method to ensure fair comparisons. The diversity criterion is selected as the termination condition, and the threshold value is optimized as 0.6. For the LaSVM, the widely used radial basis function (RBF) is selected as the kernel function, while the parameter gamma is set as the average variances of band values of labeled samples, and the parameter $C$ is optimized as 10. For the RF, according to the author's suggestion, the number of trees is set to 100, the minimum size of leaf nodes is set to 5, and the split rule is the Gini index.

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TABLE II
CONFUSION MATRIX AND ACCURACY OF THE MCSFC METHOD

| | Water | Shadows | Constructions | Bare soils | Woodlands | Grasslands | U.A. (%) |
|---|---|---|---|---|---|---|---|
| Water | 121,584 | 916 | 462 | 200 | 656 | 191 | 98.04 |
| Shadows | 938 | 15,450 | 1488 | 7 | 666 | 14 | 83.23 |
| Constructions | 55 | 77 | 82,373 | 716 | 30 | 202 | 98.71 |
| Bare soils | 0 | 0 | 132 | 2754 | 9 | 130 | 91.04 |
| Woodlands | 422 | 153 | 133 | 1003 | 49,064 | 6251 | 86.04 |
| Grasslands | 1 | 0 | 3 | 2020 | 1082 | 36,172 | 92.09 |
| P.A. (%) | 98.85 | 93.09 | 97.38 | 41.1 | 95.26 | 84.2 | |
| O.A. (%) | 94.48 | | | | | | |
| Kappa | 0.9257 | | | | | | |

In addition, four semisupervised methods, namely, the ssFCM, SKFCM-F, SIIT2-FCM, and SS-AIT2FCM algorithms, are compared. For the ssFCM and SKFCM-F algorithms, the parameter values of the fuzzifier, the maximum number of iterations, and the minimum error tolerance of the objective function are the same as those in the MCSFC method to ensure fair comparisons, and the weighting factor to maintain a balance between the supervised and unsupervised components is calculated by $1 -$ (Number of labeled samples)/(Number of samples) according to the authors' suggestion. The Gaussian kernel is used in the SKFCM-F algorithm, and the parameter of the Gaussian kernel is calculated from the distance variance of each type by the labeled samples. For the SIIT2-FCM algorithm, the two fuzzifiers $m_1$ and $m_2$ are set to 2.0 and 5.0, respectively, and the other parameters are the same as those in the MCSFC method. For the SS-AIT2FCM algorithm, the two fuzzifiers $m_1$ and $m_2$ are set to 2.0 and 5.0, respectively, and the adaptive factors for each type are calculated automatically according to the authors' suggestion.

The results are shown in Fig. 10, and the UAs, overall accuracies (OAs), and kappa coefficients are listed in Table III. Based on Table II, we find that the accuracy of the MCSFC method (OA = 94.48%) is the highest among the eight methods. Fig. 10(a) shows that the CFDT misclassifies many shadows as water and grassland as bare land, as shown in regions A and C, respectively. The LaSVM and RF have been proven to attain good classification accuracy [41], [42]; however, the LaSVM misclassifies some water regions as buildings, as shown in region B in Fig. 10(b), and misclassifies some water regions as shadows or woodland, as shown in region D in Fig. 10(b). The RF also exhibits an obvious misclassification phenomenon. For example, in region E in Fig. 10(c), some water regions are obviously misclassified as shadows.

The four FSSC methods can produce optimal centers for each cluster by their objective functions. However, the clusters corresponding to the various land cover classes usually exhibit different shapes and sizes. As a result, these so-called optimal centers do not suitably represent the clusters. From the aspect of accuracy, ssFCM, SKFCM-F, and SIIT2-FCM attain a poor ability to distinguish water and shadows, and most shadows are misclassified as water, as shown in region A in Fig. 10(d)–(f). Some constructions are misclassified as bare soil, as shown in region F in Fig. 10(d)–(f). The SS-AIT2FCM algorithm attains an improved performance on water and shadows, but obvious misclassifications remain, as shown in region D in Fig. 10(g).

As mentioned in Section II, these single-center methods are not suitable when the data of the land cover types have multiple and diverse aggregations. In this case, water data contain aggregations, and some water bodies and shadows have similar spectral attributes, leading to these single-center methods attaining a low accuracy in distinguishing water bodies from shadows. Moreover, their ability to distinguish grasslands from the woodland is also poor. The RF applies selected attributes to split the data, but the proposed method and CFDT employ all attributes to split the data into clusters. Hence, they can build trees that are more compact. However, the CFDT produces different cluster sizes and yields faults, as described in Section II-B, while the proposed approach overcomes these faults.

The runtime of the MCSFC method is compared with those of the CFDT, LaSVM, RF, ssFCM, SKFCM-F, SIIT2-FCM, and SS-AIT2FCM algorithms; the runtimes, in this case, are 150, 6, 61, 10, 399, 1101, 296, and 872 s, respectively. To ensure a fair comparison, all the methods are implemented in Java and run on the same hardware. The MCSFC algorithm is slower than the CFDT, RF, and LaSVM but faster than the four semisupervised methods. The main reason is that the CFDT, LaSVM, and RF use only the training samples to train these classifiers, and then, unlabeled samples are classified. The MCSFC method and the four semisupervised methods deal with the image and training samples directly, and the size of the image is obviously larger than the size of the training samples. Although the MCSFC method requires more computational resources, it is obviously faster than the four semisupervised classification methods and achieves higher accuracies than these methods.

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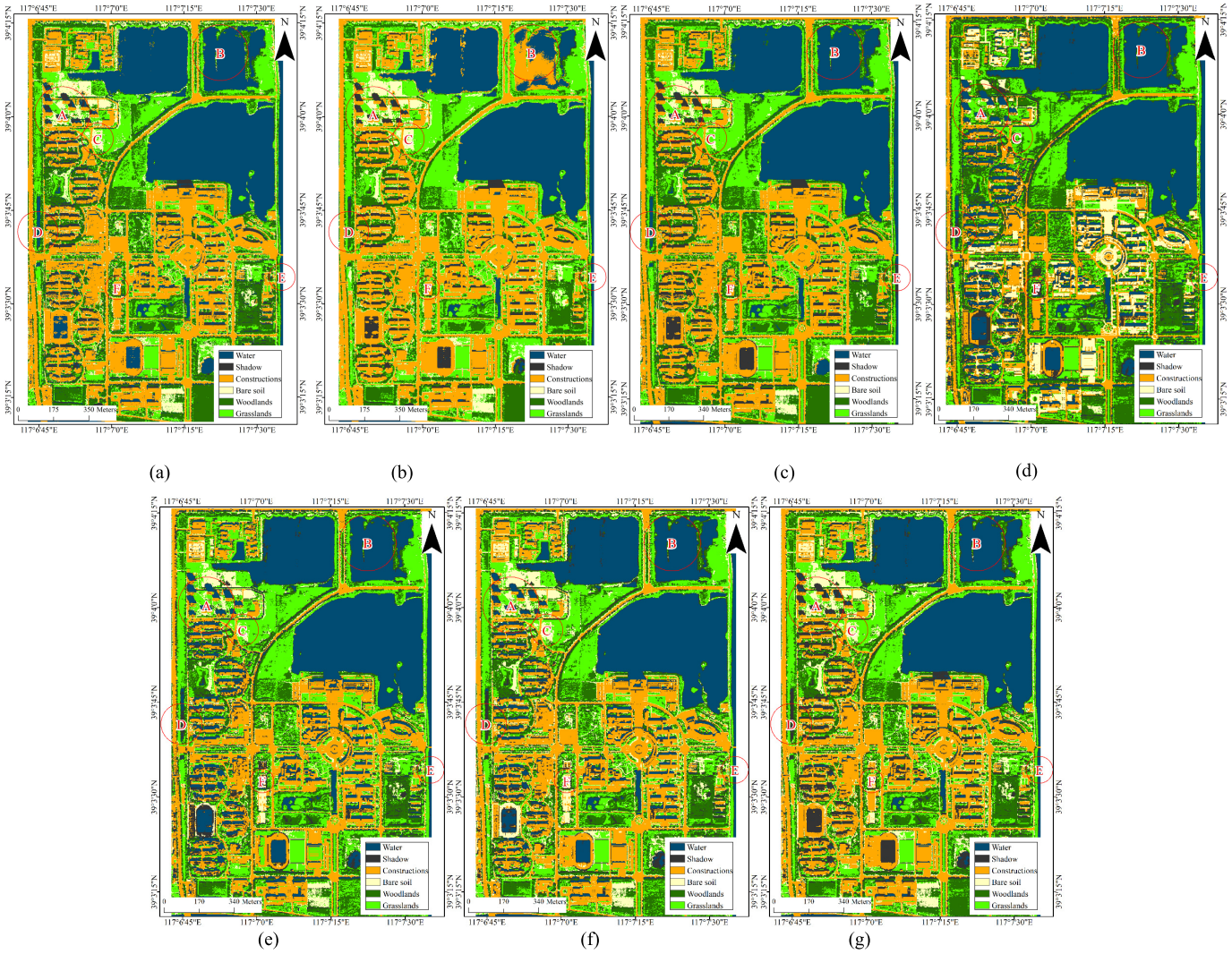


Fig. 10. Classification results of the seven methods for the WorldvView-2 dataset. (a) CFDT. (b) LaSVM. (c) RF. (d) ssFCM. (e) SKFCM-F. (f) SIIT2-FCM. (g) SS-AIT2FCM.

TABLE III
UA (%), OA (%), AND KAPPA COEFFICIENTS OF THE EIGHT METHODS

| | Proposed method | CFDT | LaSVM | RF | ssFCM | SKFCM-F | SIIT2-FCM | SS-AIT2FCM |
|---|---|---|---|---|---|---|---|---|
| Water | 98.04 | 98.17 | 89.16 | 93.61 | 98.16 | 97.73 | 95.95 | 92.46 |
| Shadows | 83.23 | 59.84 | 79.89 | 86.55 | 7.18 | 7.98 | 19.57 | 78.71 |
| Constructions | 98.71 | 97.01 | 98.7 | 98.12 | 62.25 | 82.76 | 89.36 | 93.72 |
| Bare soils | 91.04 | 92.3 | 93.91 | 89.98 | 55.02 | 87.86 | 95.48 | 95.78 |
| Woodlands | 86.04 | 82.59 | 78.37 | 87.36 | 70.4 | 62.86 | 69.75 | 74.47 |
| Grasslands | 92.09 | 78.37 | 88.62 | 90.01 | 78.98 | 87.25 | 80.33 | 80.12 |
| O.A. (%) | **94.48** | 90.51 | 89.17 | 92.8 | 76.16 | 81.29 | 83.41 | 87.38 |
| Kappa | **0.9257** | 0.8715 | 0.8558 | 0.9038 | 0.6816 | 0.747 | 0.7763 | 0.8327 |

*3) Analyzing the Effects of the Sample Volume:* This section tests the dependence of the MCSFC on sample size. The CFDT, LaSVM, RF, and SS-AIT2FCM are compared with the MCSFC, and the settings of the parameters of these methods are the same as those in the last section. Table IV lists the O.A. and the kappa coefficients of these methods with 7.7%

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TABLE IV

OA AND KAPPA COEFFICIENTS OF THE RESULTS OF THE FIVE METHODS FOR REDUCTIONS TO 7.7% AND 4.8%, RESPECTIVELY

| Sample proportion | | MCSFC | CFDT | LaSVM | RF | SS-AIT 2FCM |
|---|---|---|---|---|---|---|
| 7.7 % | O.A. (%) | **93.36** | 89.86 | 86.11 | 92.76 | 87.07 |
| | Kappa | **0.9111** | 0.8637 | 0.8162 | 0.9029 | 0.8288 |
| 4.8 % | O.A. (%) | **93.03** | 86.93 | 80.07 | 90.2 | 86.36 |
| | Kappa | **0.9063** | 0.8233 | 0.739 | 0.8955 | 0.8089 |

and 4.8% proportions of samples. It can be seen that, with the decrease in sample proportions, the accuracy losses of the MCSFC, RF, and SS-AIT2FCM are less than those of the other two methods, and the MCSFC algorithm still has the highest accuracy. Therefore, the MCSFC algorithm has good stability and reliability.

*4) Analyzing the Influences of the Granularity Size and Fuzzifier:* As discussed before, the granularity size and fuzzifier value may influence the classification results. If the granularity size is small, the data are split into many small clusters, and vice versa. In regard to the fuzzifier parameter $m$, many studies have proven that different fuzzifier values may produce different results [7], [43]–[45]. In this section, we examine the influences of the two parameters on the results.

First, the fuzzifier $m$ is set to 2.0, and the granularity size varies from 50 to 600 in steps of 50. Then, the changes in precision are analyzed. The number of labeled pure clusters, labeled impure and unlabeled clusters, and the overall accuracy (OA) are summarized in Table V. We find that different granularity sizes produce different results. In this case, when the granularity size changes from 50 to 600, the number of pure labeled and unlabeled clusters and the overall classification accuracies decrease with increasing granularity and proportion of impure labeled clusters. However, the number of impure labeled clusters increases. This implies that the ability to express the spectral diversity decreases with an increasing threshold value.

The reduction in the ability to directly express the spectral diversity of land cover types leads to a decline in the classification accuracy. Table V indicates that, when the threshold value increases from 50 to 600, classification accuracies generally show a downward trend. When the threshold value is between 50 and 100, the accuracy is 94.48%; when the threshold value is larger than 500, the accuracy is lower than 94%.

Second, we set the granularity size to 200 and change the fuzzifier $m$ from 1.8 to 4 in steps of 0.2. The number of labeled pure clusters, labeled impure clusters, unlabeled clusters, and the OA are summarized in Table VI. Although the different fuzzifiers may produce different partition results for the dataset after the hierarchical clustering process, the number of these three kinds of clusters exhibits no obvious change trend. From the aspect of accuracy, we find that the different fuzzifiers produce different results in this study, such as the FCM and FSSC methods. For this dataset, the OA is above 94% when the fuzzifier is between 1.8 and 3, and a fuzzifier between 2 and 2.2 yields the optimal results. In addition, we test the influences of parameter $C$ from 2 to 8, and the overall accuracies are very similar; thus, it is not further discussed.

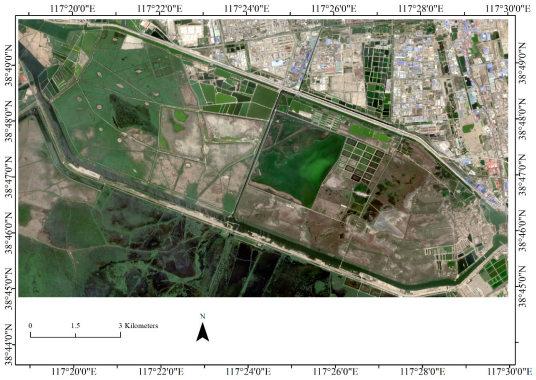


Fig. 11. Natural color composite image of the Sentinel-2 dataset using bands 4, 3, and 2.

## B. Sentinel-2 Dataset

The Sentinel-2 dataset was acquired on June 24, 2019, and it consists of 13 spectral channels in the visible, near-infrared, and short-wave infrared parts of the spectrum. Among these 13 bands, the spatial resolution of the coastal aerosol band (0.443 $\mu$m), the water vapor band (0.945 $\mu$m), and the cirrus band (1.375 $\mu$m) is 60 m. The spatial resolution of the visible bands 2–4 (spectral region ranging from 0.490 to 0.665 $\mu$m) and NIR band 8 (0.842 $\mu$m) is 10 m, and the spatial resolution of bands 5–7 (with wavelengths ranging from 0.705 to 0.783 $\mu$m), band 8 (0.865 $\mu$m), and bands 11 and 12 (1.610 and 2.190 $\mu$m, respectively) is 20 m [46]. The original Level 2A product is resampled to a 10-m resolution by the Sentinel Application Platform (SNAP) software (version 7.0) [47], and bands 2–8, 8A, 11, and 12 are selected. The test area is the northern part of the Beidagang ecological nature reserve, which is located in southern Tianjin city, China. This reserve is an important habitat zone for migratory birds in Southeast Asia, and the main land cover types include wetland, construction, and bare soil. The image size is $1629 \times 921$, and the natural color composite image using bands 4, 3, and 2 is shown in Fig. 11. When the water level of the lake notably drops in spring, floating leaf plants and submerged plants die, producing many toxins, which further cause the death of other emergent plants, such as reeds. Therefore, in this case, the dataset is classified into five types: water, construction, bare soil, green lands, and withered aquatic plants. The test samples were collected by visual interpretation and validated by using Google Earth. The strategy of the training sample selection was the same as that in experiment 1. The training samples in Fig. 12(a) cover approximately 8.6% of the test area. The test samples in Fig. 12(b) cover approximately 44.5% of the test area, and the numbers of training and test samples are listed in Table VII.

*1) Spectral Diversity and Classification Results:* In this case, parameter $C$ of the FCM is set to 3, the fuzzifier $m = 2.0$, and the number of maximum iterations is 100. The maximum

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TABLE V
NUMBER OF CLUSTERS AND CLASSIFICATION ACCURACY UNDER THE DIFFERENT GRANULARITY SIZES IN CASE 1

| Size | **50** | **100** | 150 | 200 | 250 | 300 | 350 | 400 | 450 | 500 | 550 | 600 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pure clusters (n) | 3944 | 3057 | 2597 | 2277 | 2030 | 1823 | 1673 | 1521 | 1417 | 1309 | 1207 | 1127 |
| Impure clusters (n) | 33 | 69 | 78 | 87 | 92 | 99 | 99 | 107 | 111 | 112 | 110 | 109 |
| Unlabeled clusters (n) | 21,942 | 10,395 | 6422 | 4526 | 3396 | 2700 | 2202 | 1825 | 1555 | 1336 | 1170 | 1011 |
| O.A. (%) | **94.48** | **94.48** | 94.21 | 94.44 | 94.36 | 94.06 | 94.17 | 94.09 | 94.1 | 93.88 | 93.94 | 93.93 |

TABLE VI
NUMBER OF CLUSTERS AND CLASSIFICATION ACCURACY UNDER THE DIFFERENT FUZZIFIERS IN CASE 1

| Fuzzifier | 1.8 | 2 | **2.2** | 2.4 | 2.6 | 2.8 | 3 | 3.2 | 3.4 | 3.6 | 3.8 | 4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pure clusters (n) | 2297 | 2277 | 2252 | 2232 | 2276 | 2279 | 2230 | 2270 | 2275 | 2256 | 2269 | 2240 |
| Impure clusters (n) | 95 | 87 | 94 | 92 | 108 | 109 | 107 | 105 | 100 | 99 | 96 | 95 |
| Unlabeled clusters (n) | 4202 | 4526 | 4690 | 4603 | 4742 | 4736 | 4679 | 4687 | 4670 | 4659 | 4725 | 4716 |
| O.A. (%) | 94.27 | 94.44 | **94.48** | 94.31 | 94.41 | 94.13 | 94.1 | 93.75 | 94.09 | 94.14 | 93.67 | 93.69 |

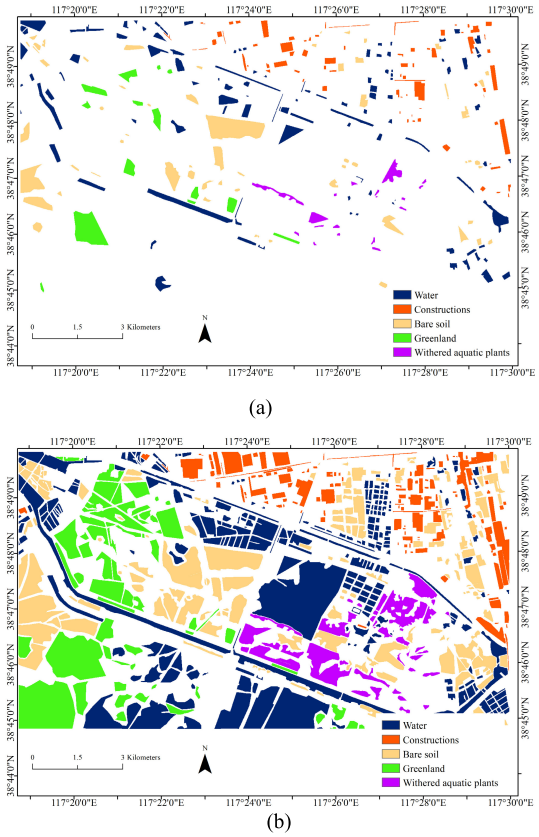


Fig. 12. (a) Training and (b) test samples for case 2.

number of samples in each leaf was set to 100. As shown in Fig. 13(a), 6322 pure labeled leaves, 1322 impure labeled leaves, and 25 616 unlabeled leaves were achieved by the hierarchical clustering process. In total, 2231, 884, 1622, 1205, and 380 leaves in these pure leaves correspond to the above five land cover types, and their spatial distributions are shown

TABLE VII
NUMBER OF TRAINING AND TEST SAMPLES IN DATASET 2

| Type | Number of training samples | Number of test samples |
|---|---|---|
| Water | 45,990 | 245,708 |
| Constructions | 17,779 | 64,826 |
| Bare soils | 40,266 | 174,418 |
| Greenland | 18,594 | 132,949 |
| Withered aquatic plants | 6845 | 49,248 |

in Fig. 13(b). It is difficult to classify the bare soil into subclasses during the labeled sample selection process, even if these samples exhibit large spectral differences. As a result, we usually select all bare soil samples as one type. For the single-center classification methods, this spectral difference is ignored because one center is adopted to express this type. In the proposed method, these spectral differences are preserved. Selecting bare soil as an example, 1622 subclusters are produced, and these subclusters preserve the spectral differences. Compared to Figs. 12(a) and 13(b), the extent of the pure leaves expands the spatial scope of the classification reference of each type. Moreover, any unreliable samples are automatically eliminated.

The classification result of the proposed method is shown in Fig. 14, and the confusion matrix and accuracy are listed in Table VIII. The U.A. of all land cover types is above 80%, and the water body attains the highest accuracy of 99.34%. The OA and the kappa coefficient are 94.6% and 0.9271, respectively.

*2) Accuracy Comparison:* In this section, the MCSFC method is compared with the three supervised methods, including CFDT, LaSVM, and RF, as well as with the four semisupervised methods: ssFCM, SKFCM-F, SIIT2-FCM, and SS-AIT2FCM. The setting of the parameters of these methods

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TABLE VIII
CONFUSION MATRIX AND ACCURACY FOR THE SENTINEL-2 DATASET OF THE MCSFC METHOD

| | Water | Constructions | Bare soils | Greenlands | Withered aquatic plants | U.A. (%) |
|---|---|---|---|---|---|---|
| Water | 244,075 | 50 | 16 | 1508 | 59 | 99.34 |
| Constructions | 31 | 54,163 | 10,558 | 11 | 63 | 83.55 |
| Bare soils | 38 | 6294 | 159,545 | 852 | 7689 | 91.47 |
| Greenlands | 5521 | 25 | 20 | 127,299 | 84 | 95.75 |
| Withered aquatic plants | 1653 | 170 | 1129 | 264 | 46,032 | 93.47 |
| P.A. (%) | 97.12 | 89.23 | 93.16 | 97.97 | 85.36 | |
| O.A. (%) | | | 94.6 | | | |
| Kappa | | | 0.9271 | | | |

TABLE IX
UA (%), OA (%), AND KAPPA COEFFICIENTS OF THE EIGHT METHODS

| | MCSFC | CFDT | LaSVM | RF | ssFCM | SKFCM-F | SIIT2-FCM | SS-AIT2FCM |
|---|---|---|---|---|---|---|---|---|
| Water | 99.34 | 98.73 | 94.11 | 99.38 | 94.84 | 98.7 | 94.71 | 96.11 |
| Constructions | 83.55 | 69.46 | 81.07 | 44.42 | 69.93 | 68.13 | 66.16 | 76.28 |
| Bare soils | 91.47 | 81.17 | 92.06 | 97.49 | 54.38 | 49.96 | 47.68 | 59.43 |
| Green lands | 95.75 | 90.64 | 98.54 | 92.21 | 5.28 | 72.66 | 71.3 | 76.66 |
| Withered aquatic plants | 93.47 | 70.03 | 88.39 | 78.84 | 88.11 | 90.56 | 94.32 | 94.29 |
| O.A. (%) | **94.6** | 87.57 | 92.77 | 90.6 | 76.37 | 77.22 | 74.95 | 80.58 |
| Kappa | **0.9271** | 0.83214 | 0.9029 | 0.8709 | 0.692 | 0.7024 | 0.6756 | 0.7464 |

is consistent with Section IV. The classification results are shown in Fig. 15, while the user and overall accuracies and kappa coefficients are reported in Table IX. The results of the four supervised classification methods are obviously better than those of the four semisupervised classification methods. Moreover, the overall accuracies of the proposed method, LaSVM, and RF exceed 90%. The proposed method attains the maximum OA and kappa coefficient among these methods. Fig. 15(a) shows that the CFDT produces high accuracies for water and green lands, at 98.73% and 90.64%, respectively, but it exhibits a poor ability to distinguish constructions and bare soils, as the outlines of the constructions are not obvious in region *A* and a large area of bare soil is misclassified into constructions in region *B*. In Fig. 15(a), region *E* reveals that the area of withered aquatic plants obtained by the CFDT is obviously smaller. In this experiment, the LaSVM attains a relatively high classification accuracy, but regions *B* and *A* in Fig. 15(b) indicate that it also exhibits a low accuracy in distinguishing constructions from bare soils, and some withered aquatic plant zones have been misclassified as bare soils in region *D*. The RF does not perform very well with this dataset, and we observe that it achieves a poor accuracy for constructions in region *A* in Fig. 15(c). However, it achieves high accuracy for water and green lands.

In this case, all four single-center semisupervised methods attain a low OA. Many bare soils are misclassified as constructions, as shown in region *B* in Fig. 15(d)–(h). Some healthy aquatic plants are also seriously misclassified as withered aquatic plants, as shown in region *C* in Fig. 15(d)–(h). Hence, the withered aquatic plant zone is greatly exaggerated.

TABLE X
OA AND KAPPA COEFFICIENTS OF THE RESULTS OF THE FIVE METHODS FOR REDUCED SAMPLE PROPORTIONS OF 6.5% AND 2.7%

| Sample proportion | | MCSFC | CFDT | LaSVM | RF | SS-AIT 2FCM |
|---|---|---|---|---|---|---|
| 6.5 % | O.A. (%) | **92.67** | 85.54 | 87.33 | 87.36 | 75.83 |
| | Kappa | **0.9003** | 0.8015 | 0.8319 | 0.8251 | 0.6868 |
| 2.7 % | O.A. (%) | **90.27** | 83.56 | 87.92 | 86.64 | 74.03 |
| | Kappa | **0.869** | 0.776 | 0.8398 | 0.8177 | 0.6633 |

*3) Analyzing the Effects of the Samples Volume:* Similar to the last section, this section tests the dependence of the MCSFC on the sample volume by reducing the test sample sizes. The CFDT, LaSVM, RF, and SS-AIT2FCM are compared with the MCSFC, and the settings of the parameters of these methods are the same as those in previous experiments. We reduced some samples to cover approximately 6.5% and 2.7% of the study area, and the OA and kappa coefficients of the results are listed in Table X. When the sample size is reduced to 6.5%, the proposed MCSFC has the highest accuracy in these methods, and the overall accuracies of these methods are reduced by 1.93%,

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TABLE XI
NUMBER OF CLUSTERS AND CLASSIFICATION ACCURACY UNDER THE DIFFERENT GRANULARITY SIZES IN CASE 2

| Size | 50 | **100** | 150 | 200 | 250 | 300 | 350 | 400 | 450 | 500 | 550 | 600 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pure clusters (n) | 7759 | 6322 | 5103 | 4218 | 3615 | 3202 | 2843 | 2548 | 2294 | 2072 | 1893 | 1773 |
| Impure clusters (n) | 854 | 1322 | 1542 | 1562 | 1537 | 1513 | 1451 | 1393 | 1338 | 1297 | 1254 | 1210 |
| Unlabeled clusters (n) | 54,238 | 25,616 | 15,588 | 10,906 | 8259 | 6464 | 5296 | 4414 | 3746 | 3258 | 2879 | 2582 |
| O.A. (%) | 94.45 | **94.6** | 93.91 | 93.88 | 93.7 | 93.26 | 93.1 | 93.01 | 93.06 | 92.96 | 92.83 | 92.65 |
| Kappa | 0.925 | **0.9271** | 0.9177 | 0.9173 | 0.9148 | 0.9088 | 0.9066 | 0.9053 | 0.9059 | 0.9046 | 0.9028 | 0.9003 |

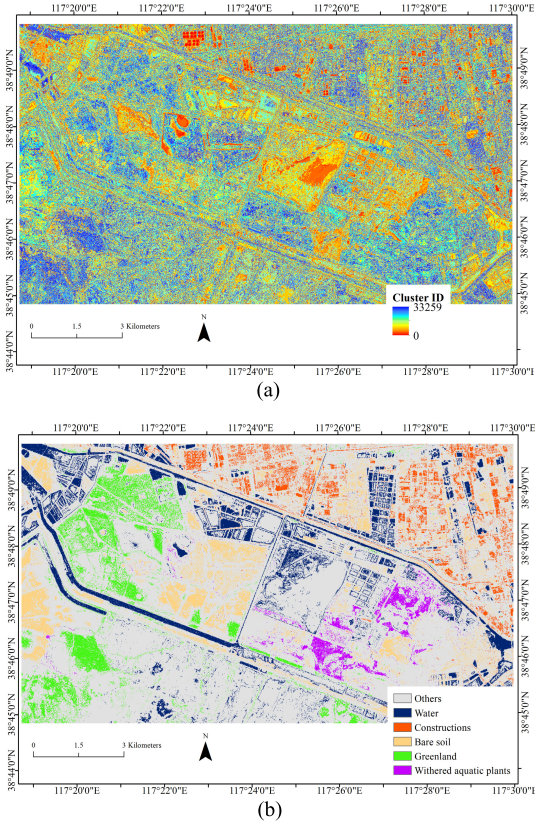


Fig. 13. (a) Full partition of image 1. (b) Spatial distribution of the pure granularities corresponding to the six types in image 2.

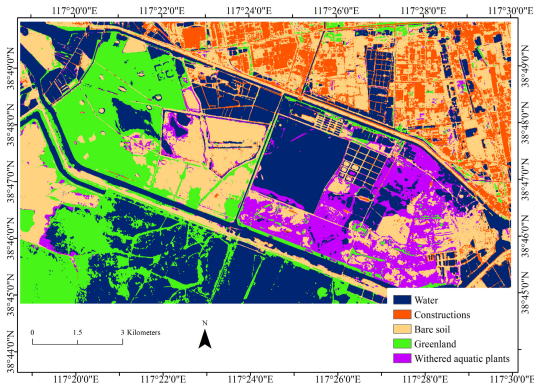


Fig. 14. Results of the MCSFC method on the Sentinel-2 dataset.

2.03%, 5.44%, 3.24%, and 4.75%. When the sample size is reduced to 2.7%, the proposed MCSFC has the highest accuracy among these methods, and it is the only method that achieves an accuracy of higher than 90%.

*4) Analyzing the Influences of the Granularities and Fuzzifier:* Similar to case 1, this section examines the influences of the above two parameters in this case. We test the influences of the granularity size by setting the fuzzifier $m$ to 2.0, and the granularity size varies from 50 to 600 in steps of 50; the number of labeled pure clusters, labeled impure clusters, unlabeled clusters, and the OA are determined. The number of labeled pure clusters, labeled impure clusters, unlabeled clusters, and the OA are listed in Table XI. We observe that the number of pure clusters has dramatically decreased, indicating that the number of centers describing the spectral differences between the ground objects has sharply decreased. Therefore, the classification accuracy decreases obviously. The accuracy decreases from 94.6% to 92.65%, and the difference between the maximum and minimum values is 1.95%. When the granularity size is 100, the accuracy is 94.6%; when the granularity size is larger than 500, the accuracy is higher than 93%.

Second, the granularity size is set to 100 because it yields the highest accuracy (see Table XI), and the fuzzifier $m$ varies from 1.8 to 4 in steps of 0.2. Then, the changes in the leaves and precision were analyzed. The numbers of labeled pure clusters, labeled impure clusters, unlabeled clusters, and the OA are listed in Table XII. We find that the number of pure granularities does not greatly change with the fuzzifier value, demonstrating that there is no obvious change in the degree of detail describing the spectral difference between the various land cover types. From the aspect of the accuracy, the highest OA is 94.71% at $m = 1.8$, and the minimum value is 94.13% at $m = 3.6$. The difference between the maximum and minimum values is only 0.58%. Therefore, in this case, although the different $m$ values produce different results, the difference between these results is very small.

It is concluded that the proposed method attains excellent performance for multispectral remote sensing images. The FCM parameter affects the performance because FCM methods with various parameter values produce different hierarchical trees and data partitions. The two cases reveal that the influence of the fuzzifier on the classification accuracy exhibits no obvious trend, and the results show that the fuzzifier influence on the accuracy is not very large; in addition, the parameter $C$ has a limited impact on the results. The

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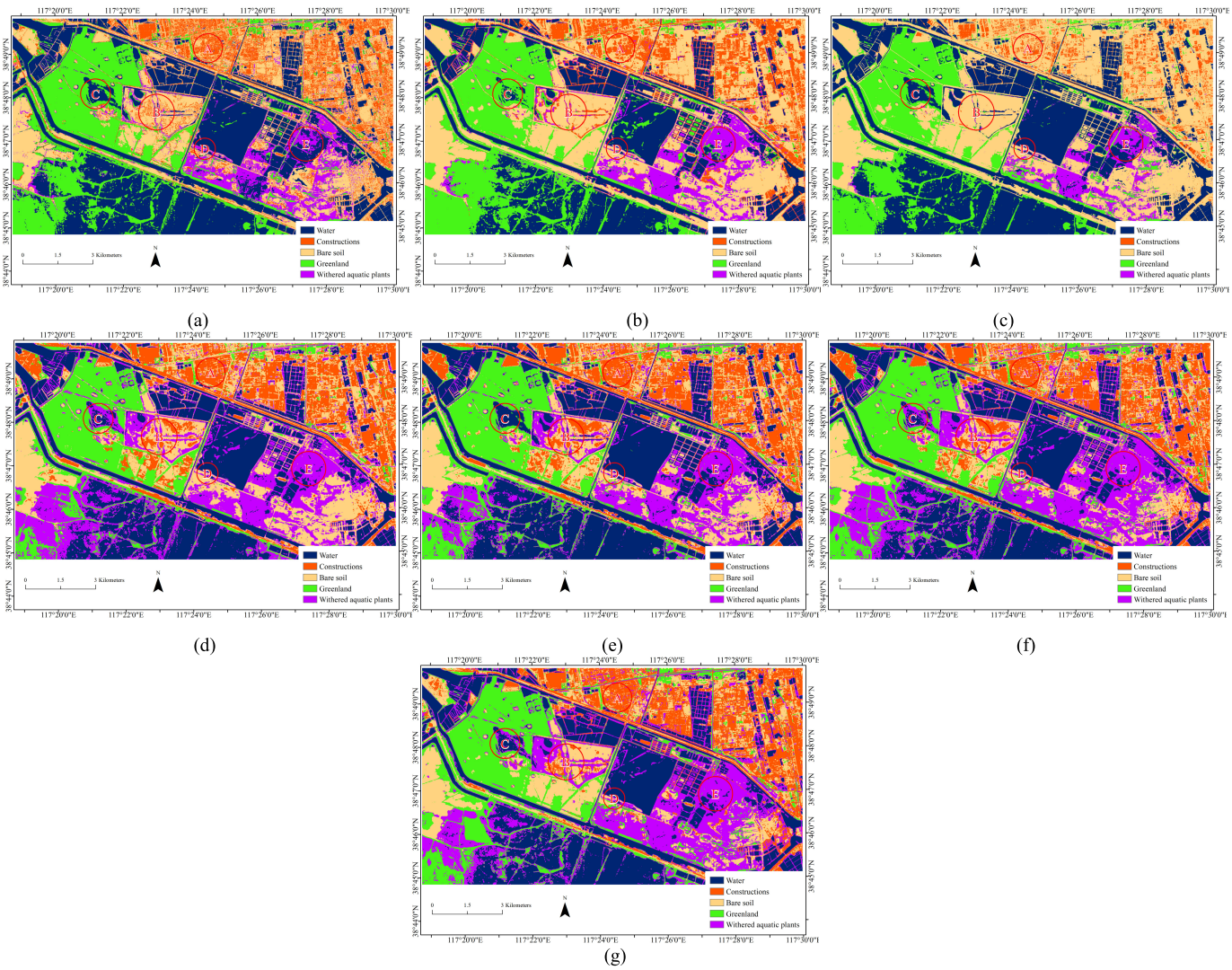


Fig. 15. Classification results of the seven methods for the Sentinel-2 dataset. (a) CFDT. (b) LaSVM. (c) RF. (d) ssFCM. (e) SKFCM-F. (f) SIIT2-FCM. (g) SS-AIT2FCM.

TABLE XII
NUMBER OF CLUSTERS AND CLASSIFICATION ACCURACY UNDER THE DIFFERENT FUZZIFIERS IN CASE 2

| Fuzzifier | **1.8** | 2 | **2.2** | 2.4 | 2.6 | 2.8 | 3 | 3.2 | 3.4 | 3.6 | 3.8 | 4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pure clusters (n) | 6374 | 6322 | 6185 | 6184 | 6217 | 6249 | 6206 | 6242 | 6151 | 6190 | 6210 | 6168 |
| Impure clusters (n) | 1368 | 1322 | 1425 | 1428 | 1418 | 1392 | 1407 | 1425 | 1410 | 1459 | 1452 | 1478 |
| Unlabeled clusters (n) | 24,435 | 25,616 | 25,767 | 26,077 | 25,934 | 25,790 | 25,774 | 25,544 | 25,706 | 25,485 | 25,484 | 25,520 |
| O.A. (%) | **94.71** | 94.6 | **94.71** | 94.17 | 94.49 | 94.3 | 94.24 | 94.42 | 94.56 | 94.13 | 94.44 | 94.36 |
| Kappa | **0.9286** | 0.9271 | **0.9286** | 0.9212 | 0.9255 | 0.9231 | 0.9221 | 0.9247 | 0.9265 | 0.9208 | 0.9249 | 0.9237 |

granularity sizes influence the ability to express the spectral diversity; that is, a small size produces a large number of granularities and a fine description of the spectral differences. Conversely, the description of the spectral difference becomes more uncertain. From the aspect of accuracy, a large size reduces the classification accuracy. Therefore, to obtain high accuracy, a small size should be adopted. In this study, the suitable granularity size in case 1 is 100 or 200, and that in case 2 is 100. In other words, different image datasets may yield different suitable granularity sizes.

## V. CONCLUSION

This study proposes the MCSFC method for multispectral remote sensing images. The comparison results indicate that the proposed method obviously improves the classification accuracy of the various FSSC methods, namely, ssFCM,

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SIIT2-FCM, SKFCM-F, and SS-AIT2FCM. Compared to the CFDT, RF, and LaSVM, the proposed method yields an obvious precision advantage. In regard to the MCSFC method, remote sensing data are split into granularities by a hierarchical clustering process, and the training samples are employed to label these granularities. The centers of the granularities that contain only one type of sample are treated as the multiple centers of the land cover types, and these centers express the spectral diversity of geographic objects. However, this objective spectral diversity is ignored by single-center classification methods. Therefore, the MCSFC method resolves the disadvantages of adopting a single center to express the spectral center of ground objects and considers this spectral diversity in the classification process. These centers are classified by the land cover type and are managed by separate cover trees. In this way, we quickly determine the shortest distance between the centers of all types and any pixel. Then, the membership degree of a pixel belonging to a given land cover type is determined by the shortest distance between this pixel and the centers of all types. The proposed method can deal with three levels of spectral diversity, and the classification accuracy is significantly improved.

Although there are several undetermined parameters in the proposed framework, such as the fuzzifier and the number of leaves, our experiments indicated that those parameters have a limited impact on the results, and the default values are recommended for these parameters in this study. The essence of the hierarchical tree is to fit the original data at different scales in multidimensional space; thus, the nodes at different levels are approximations of the data distribution. It is obvious that the fewer the samples a leaf contains, the finer the approximation, and vice versa. Therefore, regarding the granularity size, the two cases reveal that this parameter obviously affects the results. A small granularity produces a fine description of the spectral diversity and high accuracy. The experiments also show that the proposed method has good stability and reliability.

As discussed earlier, different land cover types have different sizes and forms of aggregation distribution in multidimensional space. Furthermore, the interior and boundary of these aggregations may have different spatial distributions. Therefore, how to describe these characteristics and construct accurate membership functions in the boundary zones between aggregations may have a positive influence on the modeling of the ecotone between geographical objects, which is our future work.

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